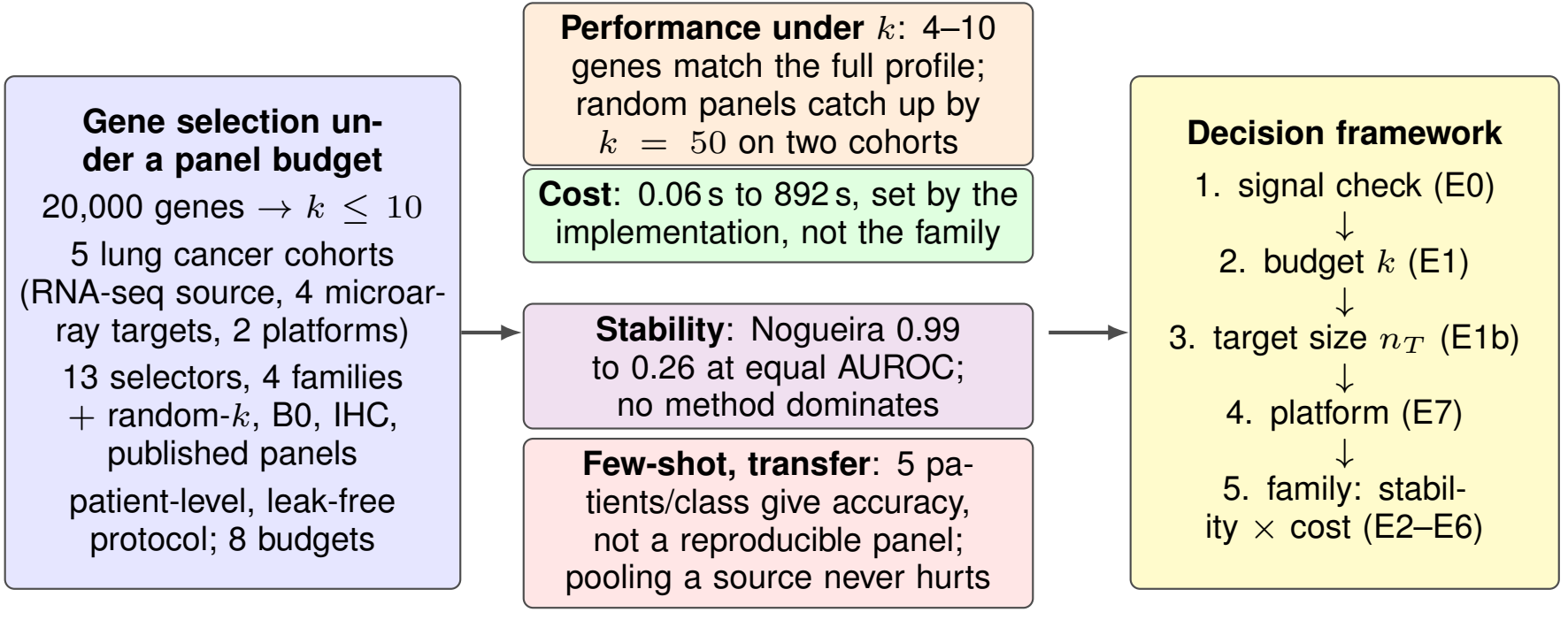

Gene selection under a panel budget
20,000 genes → $k \leq 10$
5 lung cancer cohorts (RNA-seq source, 4 microarray targets, 2 platforms)
13 selectors, 4 families + random-k, B0, IHC, published panels
patient-level, leak-free protocol; 8 budgets
Performance under k: 4–10 genes match the full profile; random panels catch up by $k = 50$ on two cohorts
Cost: 0.06 s to 892 s, set by the implementation, not the family
Stability: Nogueira 0.99 to 0.26 at equal AUROC; no method dominates
Few-shot, transfer: 5 patients/class give accuracy, not a reproducible panel; pooling a source never hurts
Decision framework
1. signal check (E0)
2. budget k (E1)
3. target size n_T (E1b)
4. platform (E7)
5. family: stability × cost (E2–E6)